

ArUco Markers in Practice

A field guide for EyeLab: printing, mounting, orienting and troubleshooting fiducial markers for AR registration in impact-hammer modal testing. Compiled July 2026 from the academic and technical sources listed at the end.

1. How a marker encodes position AND orientation

An ArUco marker is a square black border enclosing a binary grid (EyeLab uses DICT_4X4_50: a 4x4 grid, 50 unique IDs). The detector finds square contours, decodes the grid, and because the bit pattern is rotationally unique it recovers not just the ID but also which edge is the marker's TOP. Corners are always reported in a fixed order — top-left, top-right, bottom-right, bottom-left — **relative to the printed pattern, not to how you glued it**. From these four corners and the known physical size, solvePnP returns the full 6-DOF pose: the marker's local frame has X to the right, Y to the top, and Z pointing out of the printed face toward the camera.

Consequence for EyeLab: a marker glued upside-down is not 'wrong' — but the roll angle you enter in the correspondence editor must match reality. If the printed top edge points down, that is roll = 180 deg. Use the new per-marker axis overlay (AR view, 'Marker axes' checkbox) to verify: X red, Y green, Z blue. Z must point away from the surface; Y must match the roll you configured.

2. Printing

Print at 100% scale. Disable 'fit to page'. Pose scale comes entirely from the assumed physical size: a marker printed 5% small produces a 5% depth error everywhere. Measure the black square edge-to-edge with a caliper after printing and enter the measured value, not the nominal one, in the GUI marker-size field.

Matte paper only. Glossy paper glitters under lab lighting and destroys the black/white threshold locally; specular highlights are the most common cause of a marker that detects from one angle but not another (Clover docs, FoamCorePrint).

Keep the quiet zone. The detector needs a white margin around the black border — at least one module (1/6 of the marker side for 4x4 dictionaries) on every side. Do not cut along the black edge, and do not place markers on dark surfaces without a white frame.

Laser-print if possible. Inkjet blacks can be brownish and less uniform; laser toner gives a crisper edge for the sub-pixel corner refinement.

3. Flatness and mounting — your 'not super straight' problem

Paper taped to a surface is *not* flat: air bubbles, warps and curls of even 1-2 mm systematically bias the corner positions, and the pose solver amplifies that into centimetre-level depth and orientation errors. Professional calibration targets are printed on rigid substrates for exactly this reason (OKLAB, FoamCorePrint).

Practical mounting, in order of preference: (1) glue the sheet with spray adhesive onto 3-6 mm foam board, aluminium composite or glass, applying from the centre outward to push air out; (2) print stickers on matte vinyl and apply to a machined flat face; (3) if you must tape paper, tape all four edges under tension on a surface you know is flat, never over a curved or ribbed area.

Rule of thumb: out-of-plane deviation should stay below ~1% of the marker side length. For EyeLab's 12 mm markers that is 0.12 mm — effectively 'perfectly flat', which is why small markers demand rigid mounting far more than big ones.

4. Size, distance and viewing angle

Detection needs the marker to span enough pixels: ~12 px absolute minimum, 20+ px for stable sub-pixel corners (OpenCV docs; OKLAB). The projected size is $\text{size_px} = f_px \times \text{size_m} / \text{distance_m}$, with f_px the focal length in pixels (~800-1100 for a 720p-1080p webcam). A 12 mm marker at 0.5 m spans only ~19-26 px — borderline. Empirical studies find best accuracy when 100 mm markers are used between 0.35 m and 1.25 m (CEUR-WS 2744, short14).

Marker side	Reliable detection*	Stable pose*
12 mm	< 0.7 m	< 0.4 m
25 mm	< 1.4 m	< 0.9 m
50 mm	< 2.8 m	< 1.8 m
100 mm	< 5.5 m	< 3.5 m

*Indicative for a 1080p webcam (f approx 1000 px), good light, near-frontal view. Halve the ranges for 720p or oblique views.

Viewing angle: accuracy degrades steadily past ~50-60 deg off-normal, and very shallow views both lose detection and trigger the planar-pose ambiguity below. Try to keep the camera within ~45 deg of each marker's normal during registration.

5. The planar pose ambiguity — why one marker 'flips'

A single planar square seen in perspective admits **two** geometrically valid poses (mirror solutions about the image plane); this is a property of the problem, not a bug, and no PnP solver can resolve it from one view alone (Collins & Bartoli, IPPE; OpenCV forum). When the marker is small, distant or near-frontal, image noise makes the solver hop between the two solutions — the familiar Z-axis flip / jitter.

Mitigations, all of which EyeLab now applies: solve with IPPE to get both candidate poses explicitly; pick the candidate consistent with a temporal prediction (Kalman prior); gate low-marker-count poses against translation/rotation jumps; and prefer multi-marker solves whenever more than one marker is visible, because markers on different planes break the ambiguity entirely.

This is why acquisition still wants 3 markers even though a single marker can now SUSTAIN the lock: with no prior pose there is no way to pick the right one of the two single-marker solutions reliably.

6. Multi-marker layout on the structure

Placement quality matters more than marker count beyond ~4-5 markers (arXiv 2509.17345). Guidelines: **spread markers wide** — registration error scales with the ratio of structure size to marker-cloud size; **avoid collinear layouts** — EyeLab's condition number warns about this; **use different orientations** — markers angled to each other (e.g. on perpendicular faces of the L-flangia) improve pose accuracy and stability markedly over coplanar sets (Oscadal et al., Sensors 2020); **measure positions to the marker CENTRE**, in the UNV frame, as precisely as you place accelerometers; a 1 mm error in a correspondence is a 1 mm floor on registration RMS.

In structural-dynamics work the reference approach is Cepen et al. (Experimental Techniques 2024): ArUco dodecahedrons on the hammer and on the structure, poses averaged over all visible faces, bad detections rejected by reprojection distance. Their conclusions transfer directly: multiple angled markers + outlier rejection + averaging is what makes marker tracking metrologically usable.

7. Camera calibration

Pose accuracy is bounded by calibration quality (CEUR-WS 2744). Use the ChArUco wizard; capture 15+ views that cover the whole sensor including corners, at varied tilts; keep the board rigid (glass or foam

board); aim for RMS below ~ 0.5 px and recalibrate whenever resolution, zoom or focus changes. The board must also be measured after printing — EyeLab's 'Board spec' dialog lists the expected square sizes.

8. Lighting and motion

Even, diffuse light beats bright light. Avoid: direct spotlights or windows creating specular patches on the marker; hard shadows across the border (the adaptive threshold usually copes, but contrast at the border is what drives corner precision); motion blur — at typical webcam exposure, keep relative motion slow during registration, or raise shutter speed / add light. If detection is intermittent, inspect the FLT tab: the CLAHE-enhanced image shows what the detector actually sees.

9. Pre-session checklist

#	Check	Failure it prevents
1	Marker measured with caliper; size entered in GUI	global scale/depth error
2	Matte print, quiet zone intact	angle-dependent detection loss
3	Marker glued flat on rigid backing (<0.1 mm bow for 12 mm)	systematic pose bias
4	Roll/normal in correspondence editor match physical placement	wireframe rotated/mirrored
5	Marker axes overlay checked: Z out of face, Y = configured top	silent 180-deg roll errors
6	Markers spread wide, not collinear, ideally on 2+ planes	ill-conditioned registration
7	Camera within ~45 deg of marker normals at working distance	ambiguity flips, poor corners
8	Calibration RMS < 0.5 px at the session resolution	curved/offset overlay
9	No specular glare on markers from the working viewpoint	intermittent detection
10	3+ markers visible once for lock acquisition	no initial pose

Sources

- Garrido-Jurado et al., "Automatic generation and detection of highly reliable fiducial markers under occlusion", Pattern Recognition 47(6), 2014 (the original ArUco paper).
- Cepon, Ocepek, Kodric, Demsar, Bregar, Boltezar, "Impact-Pose Estimation Using ArUco Markers in Structural Dynamics", Experimental Techniques 48:369-380, 2024. <https://link.springer.com/article/10.1007/s40799-023-00646-0>
- Oscadal et al., "Improved Pose Estimation of Aruco Tags Using a Novel 3D Placement Strategy", Sensors 20(17):4825, 2020. <https://www.mdpi.com/1424-8220/20/17/4825>
- Collins & Bartoli, "Infinitesimal Plane-based Pose Estimation" (IPPE). <https://github.com/tobycollins/IPPE>
- "Investigation of ArUco Marker Placement for Planar Indoor Localization", arXiv:2509.17345. <https://arxiv.org/pdf/2509.17345>
- "Modeling ArUco Markers Images for Accuracy Analysis of Their 3D Pose Estimation", CEUR-WS Vol-2744. <https://ceur-ws.org/Vol-2744/short14.pdf>
- OpenCV documentation, "Detection of ArUco Markers" and "Calibration with ArUco and ChArUco". https://docs.opencv.org/4.13.0/d5/dae/tutorial_aruco_detection.html
- OKLAB, "ChArUco Calibration Boards: Complete Guide". <https://www.oklab.com/blog/charuco-calibration-boards-complete-guide-to-professional-camera-calibration>
- Clover documentation, "Fiducial markers (ArUco)" - practical printing advice. <https://clover.coex.tech/en/aruco.html>
- FoamCorePrint, "ChArUco Targets Boards" - substrate and matte-finish guidance. <https://www.foamcoreprint.com/charuco-targets>

Note: quantitative ranges in section 4 are extrapolated from the cited studies (which used 100 mm markers and industrial cameras) via the pinhole model; treat them as planning estimates, not guarantees.